

E-commerce dataset by Olist

A Brazilian E-commerce Giant

1. Executive Summary:

This project analyzes the Olist Brazilian E-commerce dataset to generate meaningful insights into sales performance, customer purchasing behavior, logistics efficiency, product demand, and payment trends. By leveraging SQL for data querying, Power BI for dashboard development, and Python for advanced analysis, the project aims to support data-driven business decisions. The outcome will help identify growth opportunities, improve customer satisfaction, and optimize operational processes.

2. Problem Statement:

Background: In a competitive e-commerce environment, businesses face challenges such as delayed deliveries, declining customer satisfaction, inconsistent sales across categories, and difficulty retaining repeat customers. Without proper analysis, these issues can impact profitability and long-term growth.

Objective: To analyze historical e-commerce transaction data and uncover patterns related to revenue generation, customer behavior, delivery performance, payment preferences, and product category performance.

Scope: The analysis will focus on order transactions, customer data, payments, product categories, seller performance, and customer reviews across Brazil using the Olist dataset.

3. Data Sources:

Primary Data: <https://www.kaggle.com/datasets/terencicp/e-commerce-dataset-by-olist-as-an-sqlite-database/data>

Secondary Data: Publicly available Olist Brazilian E-commerce dataset sourced from Kaggle. The dataset includes multiple relational tables such as customers, orders, order items, products, sellers, payments, reviews, and geolocation data.

4. Methodology:

Data Collection: Download and import the Olist dataset files from Kaggle into SQL Server / MySQL / PostgreSQL and Power BI.

Data Preparation:

- Handle missing and null values
- Remove duplicates
- Standardize column formats
- Convert date fields into usable formats
- Build relationships between tables

Analysis Techniques:

- Descriptive analytics
- Trend analysis
- Customer segmentation
- Delivery performance analysis
- Revenue and profitability analysis
- Review sentiment analysis
- Repeat customer analysis

Tools:

- SQL for data extraction and joins
- Power BI for dashboard creation
- Excel for quick preprocessing
- Python (Pandas, Matplotlib, Seaborn)

5. Expected Outcomes:

- Interactive dashboard showing revenue, orders, customers, and product trends
- Insights into top-performing categories and regions
- Analysis of delayed deliveries and operational inefficiencies
- Identification of repeat customer behaviour and customer satisfaction trends
- Recommendations for improving logistics, retention, and sales performance

6. Risks and Challenges:

- Data quality issues may require additional cleaning.
- Inaccurate predictions could have real-world consequences; validation is essential.
- Missing values in delivery or review data may impact analysis quality
- Data is from Brazil, so currency/language translation may require preprocessing
- Inconsistent category naming may need standardization
- Large dataset size may require optimization in Power BI model design

7. Conclusion:

This project provides a valuable opportunity to analyze a real-world e-commerce business dataset using modern analytics tools. The findings can support strategic decisions in sales growth, customer experience, logistics management, and operational efficiency. By transforming raw data into actionable insights, the project demonstrates practical business intelligence and analytics capabilities relevant to today's job market.